

Project Report

Of

NEXGENPLAYZ

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BCA-5

Under The Guidance of

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Acknowledgement

The project on **ONLINE GAME SELLING STORE** is developed in **PHP** Language at **GEETANJALI COLLEGE OF COMPUTER SCIENCE & COMMERCE**.

I would like to acknowledge that my project has been completed and I am ensuring that, in this accomplishment, I would like to express my special gratitude to all my teachers and specially to **PROF. HARSH JOSHI** without their guidance and feedback, it is not possible to complete this assignment.

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INTRODUCTION

Abstract

NexGenPlayz is a digital platform for selling video games, developed using **PHP** and **MySQL**. The system's primary goal is to provide a seamless and secure alternative to traditional retail for game distribution. Key features include full **User Authentication** (login/register), a dynamic **Game Catalogue** that fetches real-time inventory and pricing, and a dedicated **User Profile** where customers can access their purchased game library and manage their account details. The project successfully implements an end-to-end e-commerce solution for digital game assets, optimizing the distribution process and enhancing the user experience.

Purpose

The main reason we made the NexGenPlayz Online Game Selling Store was to create a simple, modern way for people to buy video games.

The goals of this project are:

>To Solve the Problem of Waiting: Unlike buying physical discs, this online system lets users buy and get their games instantly without going to a shop.

>To Keep Track of Games: The system is needed to properly manage all the games we sell (inventory) and to record which games each user owns.

>To Make it Easy: To give customers a simple website to log in, find games, pay, and get help.

In short, this project is a complete online shop made to make selling and buying games easier, faster, and better managed.

Project Profile

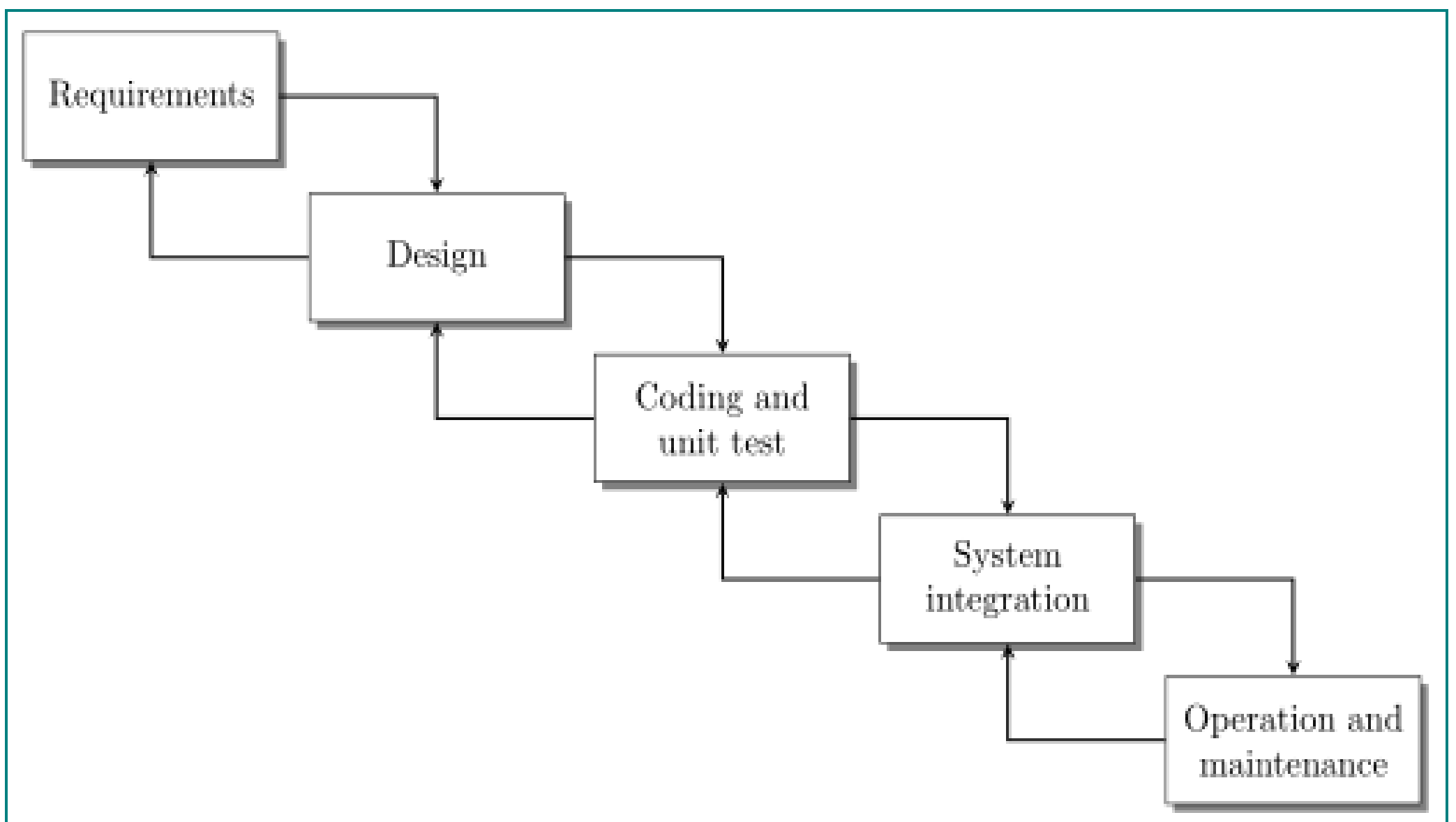
Project Title	NexGenPlayz: Online Game Selling Store
Project Type	Web Application (E-Commerce)
Front-End Technology	HTML5, CSS, JavaScript
Back-End Language	PHP
Database Used	MySQL
Development Tools	Visual Studio Code

SDLC Process Model

The Process Model used in our project **NEXGENPLAYZ** is waterfall model.

SDLC Process Model:

The **Waterfall Model** was selected as the Software Development Life Cycle (SDLC) model for the development of the NexGenPlayz system. This model is ideal because the project requirements (such as user login, game inventory management, and purchasing flow) were clearly defined and stable from the beginning of the project.



➤ **Reason the waterfall model in the software development cycle:**

- Since we have well known, clean and fixed requirements therefore its best suits for our software development.
- Our product definitions are stable.
- Technology is clearly understood.
- The project is short.

➤ **Advantages of the waterfall model:**

- This model is simple and easy to understand and use. In this model phases are processed and completed one at a time and phases do not overlap.
- Waterfall model works well for smaller projects where sequence is very well understood

Software Resources

The following software and resources were essential for building and running the **NexGenPlayz Online Game Selling Store**. These tools handle everything from writing the code to managing the database.

Category	Software/Resource Name	Purpose in Project
Back-End Language	PHP	Used for all server-side logic, database interactions (fetching games, processing forms), and security handling.
Database System	MySQL	The relational database used to store all structured data, including users, games, support messages, and payments.
Server Stack	XAMPP/WAMP	Provides the necessary local environment (Apache web server and MySQL) to run the PHP and database files during development.
Code Editor	Visual Studio Code	The main software used to write, manage, and edit all the project's source code files (PHP, HTML, CSS, SQL).

Client-Side	HTML5, CSS3, JavaScript	Used to create the structure, styling, and interactive features on the website seen by the customer
Libraries/Tools	Boxicons	A library used to easily add professional icons (like the profile and time icons) throughout the website pages.

List Modules

The NexGenPlayz system is logically divided into these five distinct modules. Each module handles a specific set of functions and interactions within the online store

- **Authentication Module:** This handles the security and access control...
- **Game Catalogue Module:** This is the storefront where all games are displayed...
- **User Profile Module:** Once a user logs in, this module allows them to view their personal account details...
- **Support Module:** This module provides customer service...
- **Admin Module:** This module is responsible for administrative tasks...

Data Dictionary

Users Table (Customer Accounts):

Column Name	Data Type	Constraint/Description	Role in Application
user_id	INT	Primary Key, Auto-Increment	Unique identifier for every user.
username	VARCHAR (50)	Not Null, Unique	Displayed name on the user profile.
email	VARCHAR (100)	Not Null, Unique	Used as the login ID for users.
Password hash	VARCHAR (255)	Not Null	Stores the secure, encrypted password.

Games Table (Store Inventory):

Column Name	Data Type	Constraint/Description	Role in Application
game_id	INT	Primary Key, Auto-Increment	Unique identifier for every game.
Title	VARCHAR (100)	Not Null	The name of the game displayed on Games.php.
price	DECIMAL (10, 2)	Not Null	The price of the game listed for purchase.
video_preview	VARCHAR (255)	Not Null	Path to the game's cover art.

cover_image	VARCHAR (255)	Not Null	Path to the game's cover art.
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User owned games Table (Purchase Record):

Column Name	Data Type	Constraint/ Description	Role in Application
ownership_id	INT	Primary Key	Unique identifier for a single ownership record.
user_id	INT	Foreign Key	Links the record to the customer who made the purchase.
game_id	INT	Foreign Key	Links the record to the specific game purchased.
purchase_date	DATETIME	Records the time of purchase.	Tracks when the game was added to the library.

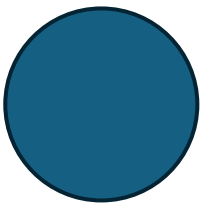
Support messages Table (Customer Queries):

Column Name	Data Type	Constraint/ Description	Role in Application
message_id	INT	Primary Key	Unique identifier for the message.
name	VARCHAR (100)	Not Null	Sender's name from the Support.php form
email	VARCHAR (100)	Not Null	Sender's contact email.

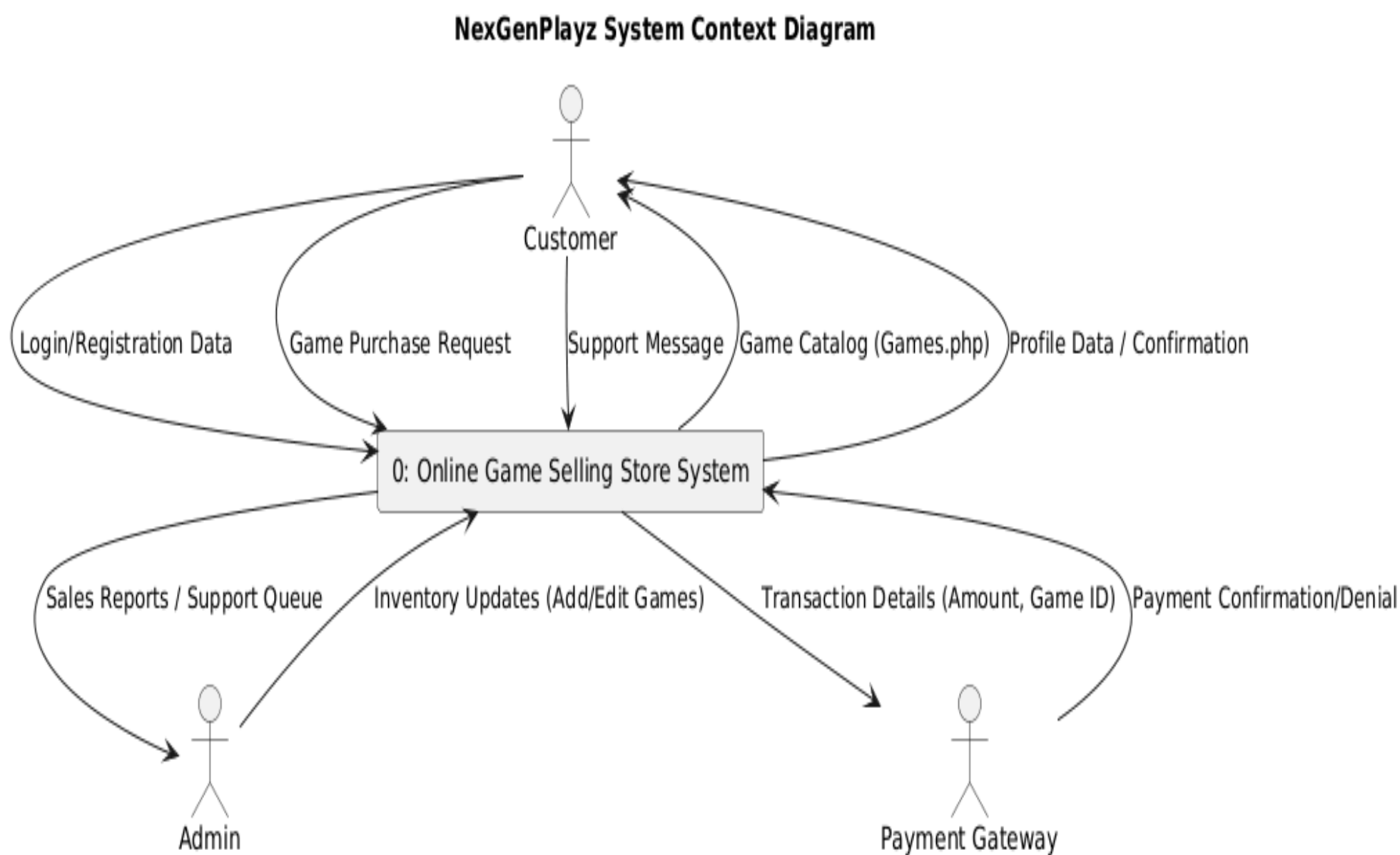
subject	VARCHAR (255)	Not Null	The topic of the support request.
message	TEXT	Not Null	The full text of the query.

DFD [Data Flow Diagram]

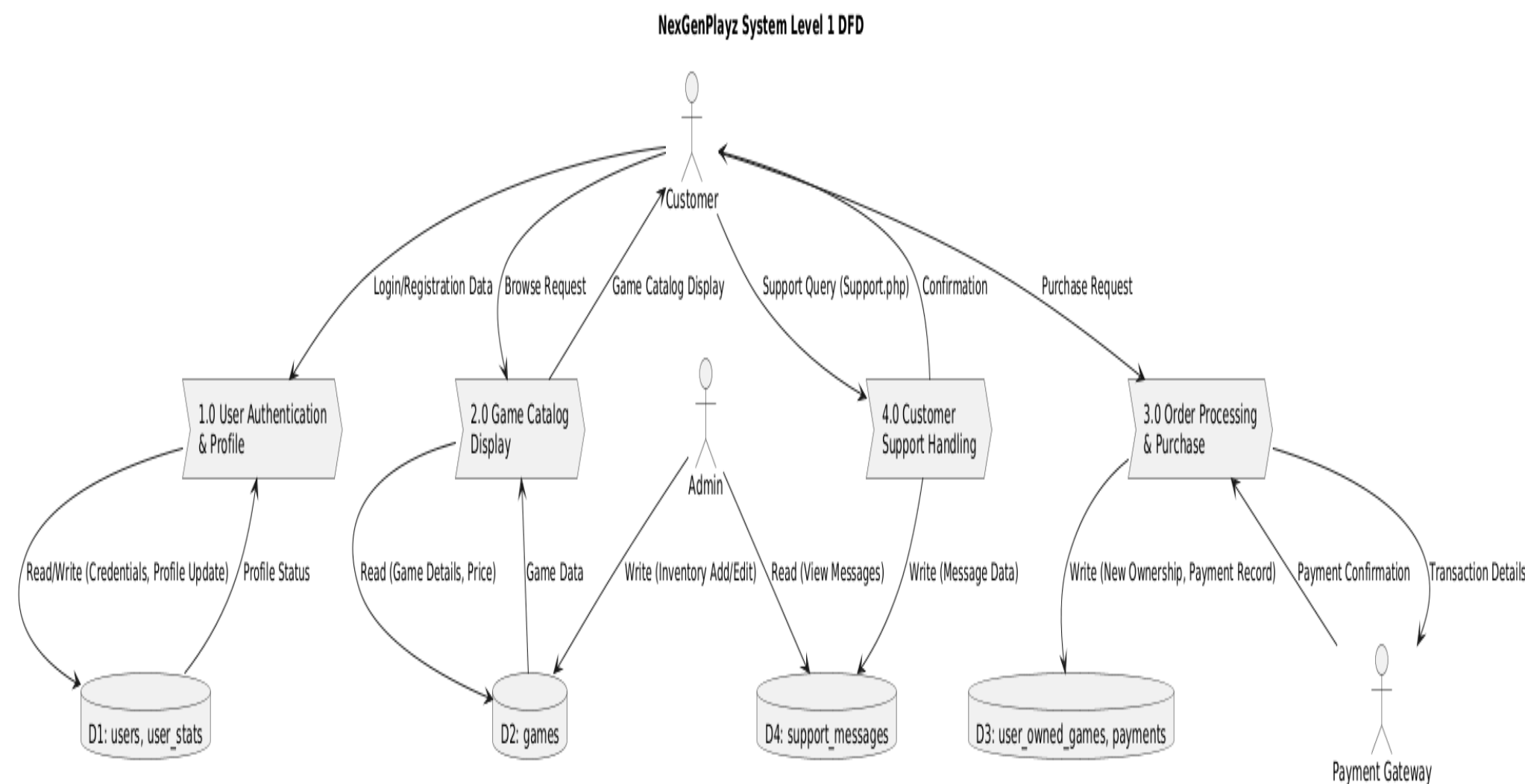
Data flow diagram symbol

Symbol	Description
	Data Flow: Data flow are pipelines through the packets of information flow.
	Process: A Process or task performed by the system.
	Entity: Entity Are object of the system. A source or destination data of a system.
	Data Store: A place where data to be stored.

0 level DFD (Context Diagram):

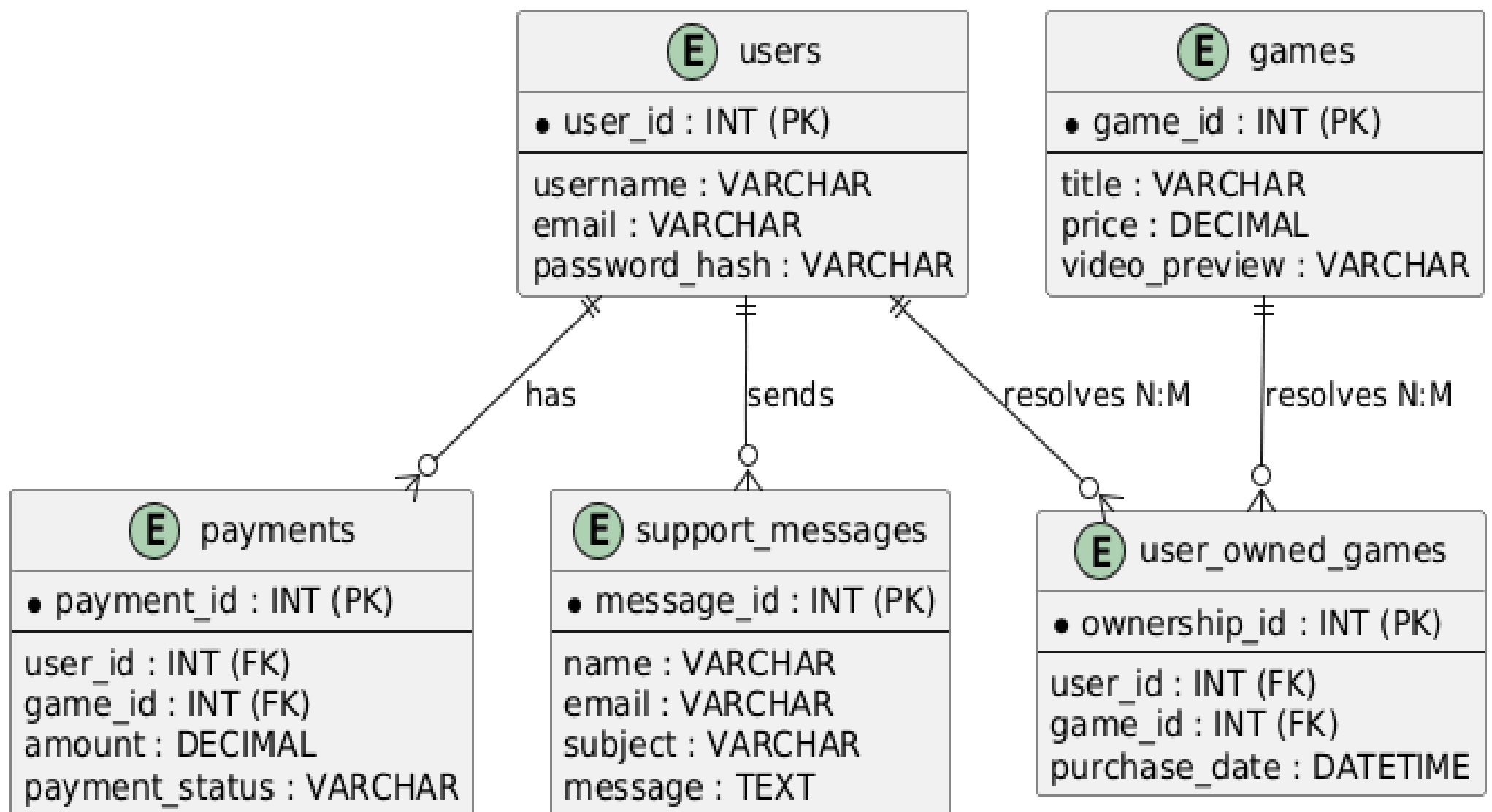


Level 1 DFD (Context Diagram):



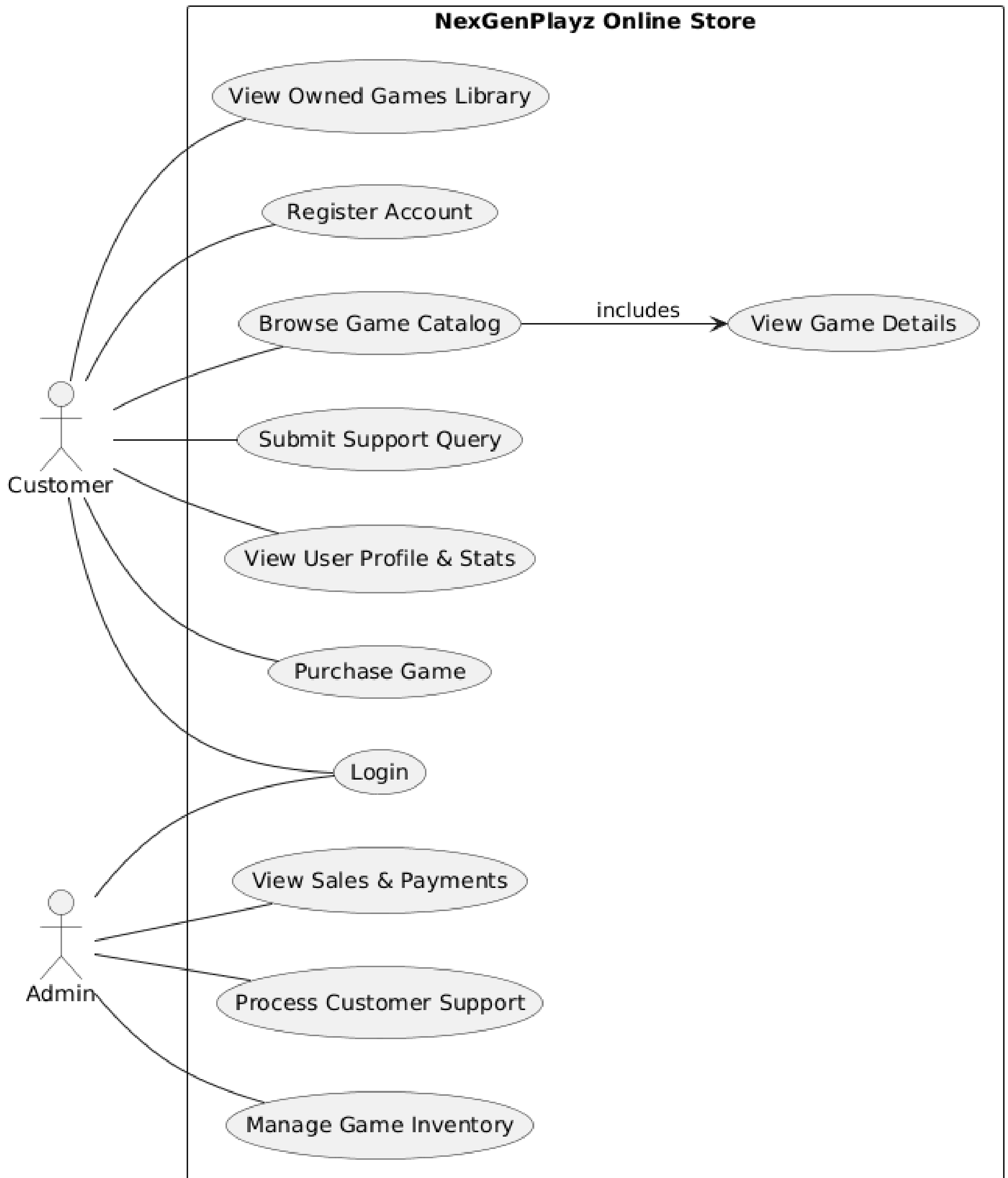
ERD [Entity Relationship Diagram]

NexGenPlayz E-R Diagram



Use Case Diagram

NexGenPlayz Use Case Diagram



Testing – Test Cases



User Login & Registration Tests:

ID	Goal	Test Steps	Expected Outcome
R-1	Successful New Account	1. Go to the Register form. 2. Enter a unique username, a unique email, and matching passwords 3. Click "Register".	The user is redirected to the Login page.
R-2	Passwords Do Not Match	1. Go to the Register form. 2. Enter "testpass" in the Password field. 3. Enter "differentpass" in the Confirm Password field. 4. Click "Register".	The system shows an error message: "Error: Passwords do not match."
R-3	Duplicate Email Check	1. Register a new user successfully (R-01). 2. Immediately try to register a second user using the exact	The system shows an error message: "Error: This email is already registered..."

		same email address. 3. Click "Register".	
R-4	Missing Fields	1. Go to the Register form. 2. Fill in the Username and Email, but leave the Password fields empty. 3. Click "Register".	The system shows an error message: "Please fill all required fields."
L-5	Successful Login	1. Register a user (R-01). 2. Go to the Login form. 3. Enter the registered email and the exact plain password 4. Click "Login".	The user is redirected to the Home.php page, and the session is started (loggedin is true).
L-6	Incorrect Password	1. Go to the Login form. 2. Enter a registered email but enter a wrong password (e.g., "wrongpass"). 3. Click "Login".	The system shows an error message: "Invalid email or password."
L-7	Non-existent Email	1. Go to the Login form.	The system shows an error message:

		2. Enter an email address that is not in the user database. 3. Enter any password. 4. Click "Login".	"Invalid email or password."
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➤ Profile Management Tests

ID	Goal	Test Steps	Expected Outcome
1	View Profile Data	1. Log in. 2. Navigate to Profile.php.	The page displays the correct username, profile picture, and the list of owned games for the logged-in user.
2	Change Username	1. Go to edit_profile.php. 2. Change the Username field to a new name. 3. Click "Save Changes".	A success message appears, and the new username is saved and displayed on Profile.php.
3	Change Profile Picture	1. Go to edit_profile.php. 2. Upload a small image file	The old picture is replaced, and the new picture

		(JPG or PNG) using the "Change Photo" option. 3. Click "Save Changes".	is displayed on Profile.php.
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➤ Game Purchase Tests:

ID	Goal	Test Steps	Expected Outcome
1	View Game List	1. Log in and go to Games.php.	The page loads all games from the database, showing the title, video preview, and price for each one.
2	Start Purchase	1. Go to Games.php. 2. Click the price button for any game	The user is taken to the PaymentGateway.php page for that specific game.
3	Successful Purchase	1. Go to the PaymentGateway.php for a game you do not own. 2. Fill in the required payment details. 3. Click the "Pay" button (which posts to process_purchase.php).	The user is sent to purchase_success.php with a message like "Purchase successful!..." The game now appears on their Profile.php page.
4	Already Owned Game	1. Attempt to purchase a game that the user already owns.	The purchase process should recognize that the game is already owned, and no duplicate purchase occurs.

➤ Support & Contact Tests:

ID	Goal	Test Steps	Expected Outcome
1	Send Support Message	1. Go to Support.php. 2. Fill in the Name, Email, Subject, and Message fields completely. 3. Click "Send Message".	A success message is displayed (e.g., "Thank you! Your message has been sent..."). A new message entry is created in the database.
2	Submission Fails (Missing Field)	1. Go to Support.php. 2. Fill in the form, but leave one field (e.g., Subject) empty. 3. Click "Send Message".	The system shows an error message like "Please fill out all fields."

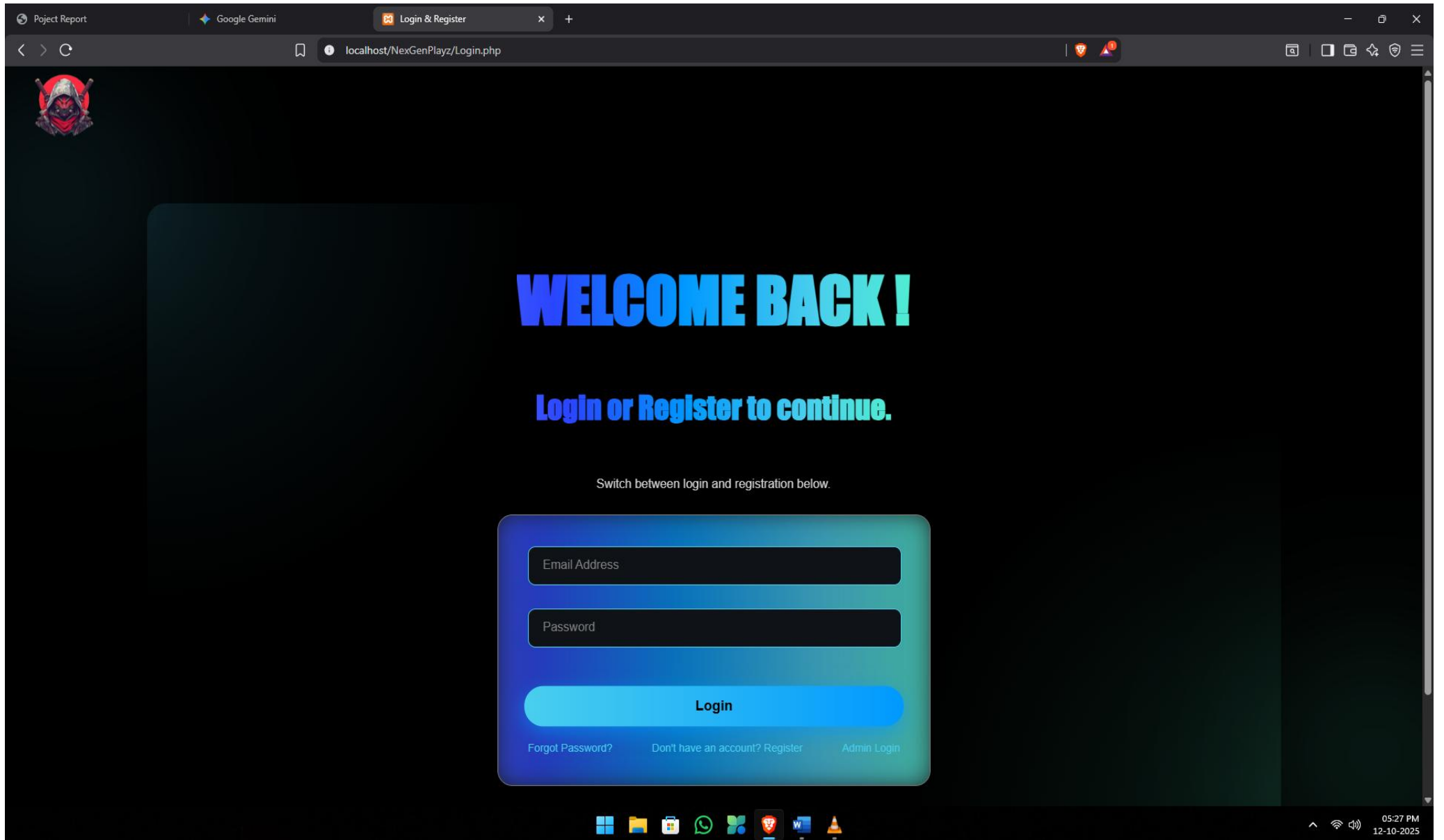
➤ Admin Functionality Tests:

ID	Goal	Test Steps	Expected Outcome
1	Admin Login	1. Go to Admin_Login.php. 2. Enter valid admin credentials.	The admin dashboard appears, showing the options to

		3. Click "Log In".	manage games and support messages.
2	Add New Game	1. Log in as Admin. 2. Use the "Add New Game" form to submit a new title, price, and video link.	The game is added to the database. The new game appears on the public Games.php page.
3	Resolve Support Ticket	1. Log in as Admin. 2. Find a "Pending" support message. 3. Click the "Resolve" button next to it.	The message disappears from the Admin dashboard, and its status in the database is updated to "resolved".

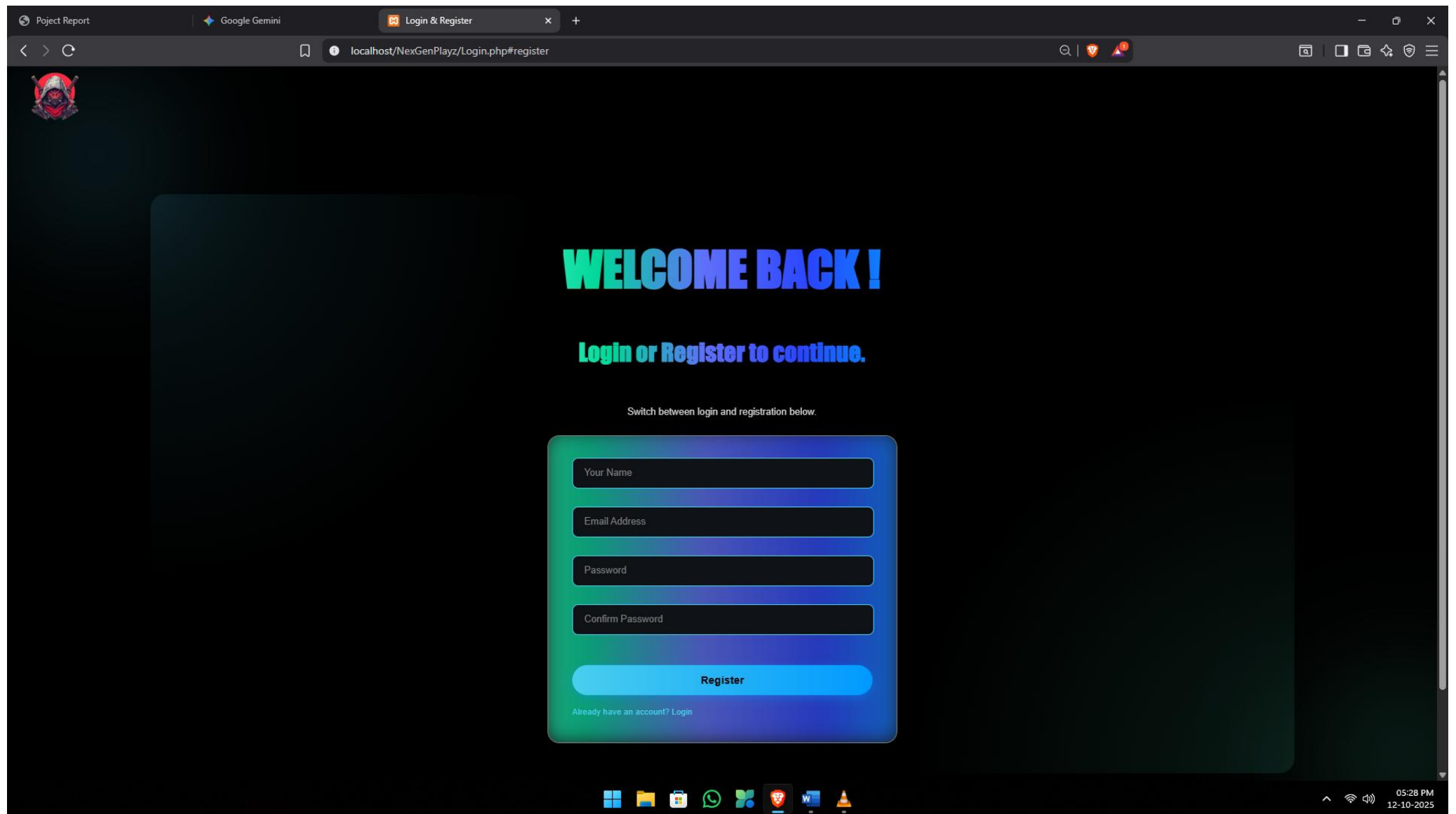
Snapshots

1. Login Page:



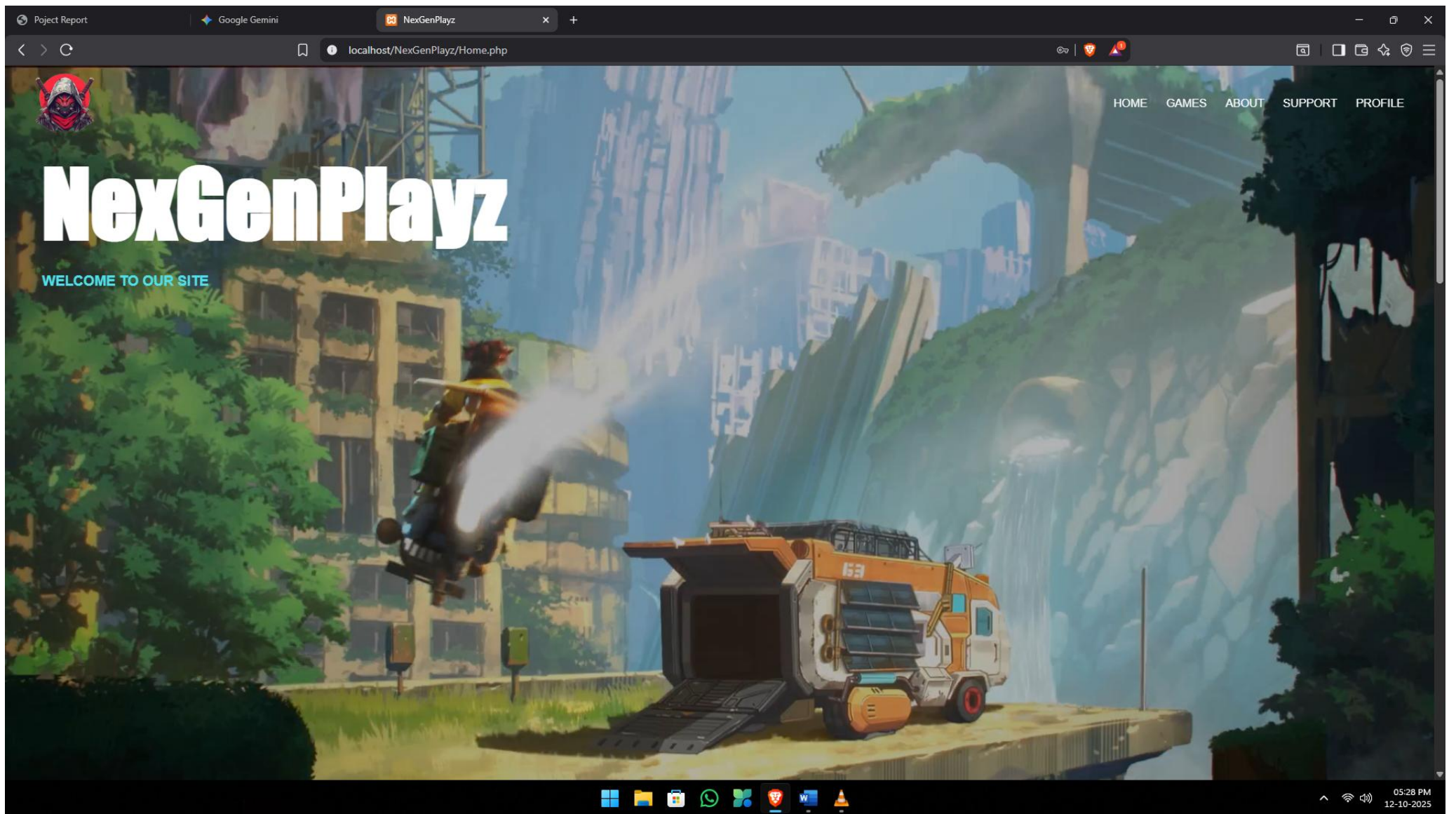
- Login page for user & Admin login.
- User needs to add their email and password to login.

2. Register Page:



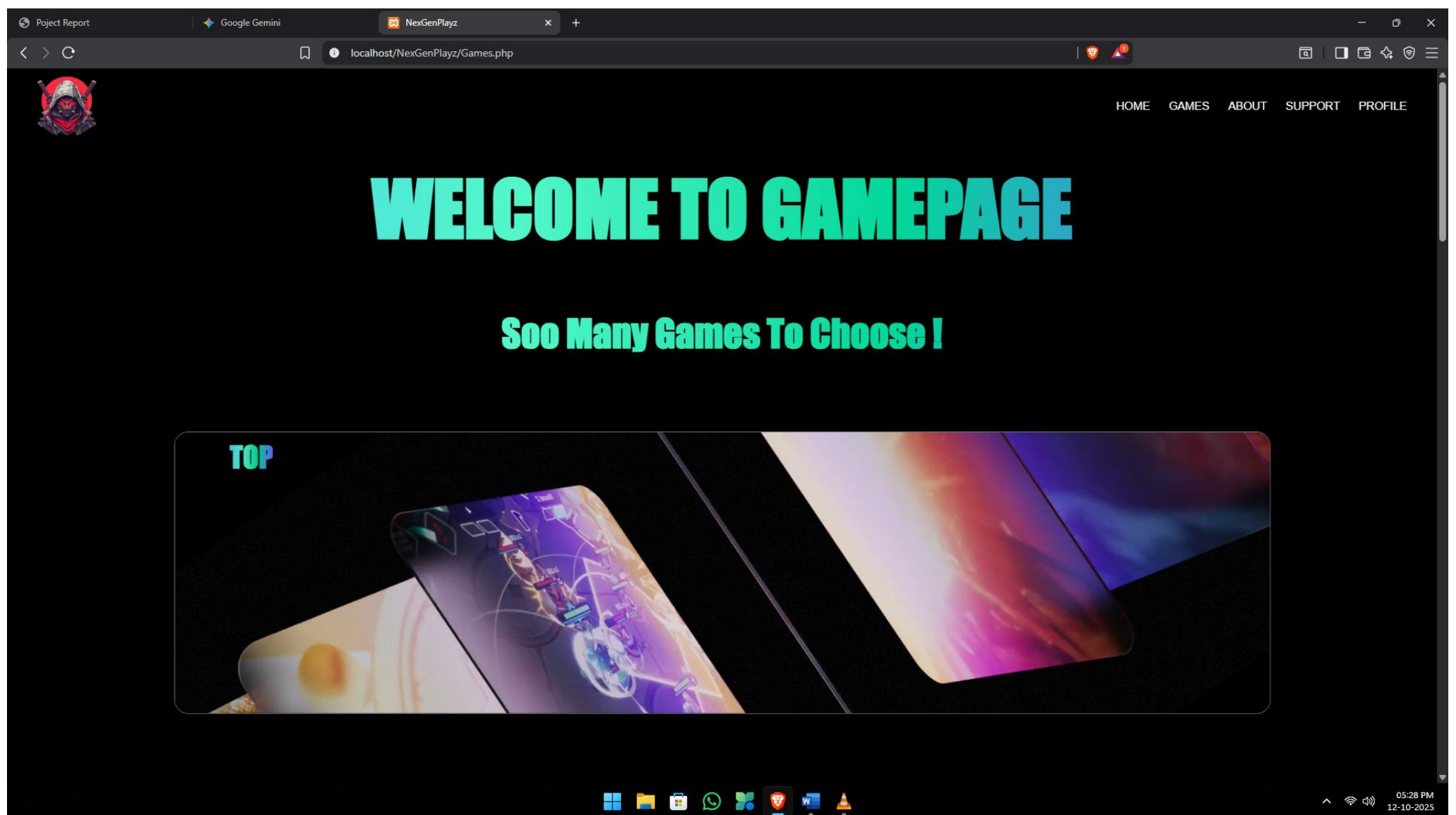
- First User Need to Register Their Account.
- After Register Their Account the User Is Redirects to Login Page.
- If The User Enters Any Wrong Information the Error Message Is displays Like “Invalid Email or Password.”
- The Same Email is Not Registered Second Time.

3. Home Page:



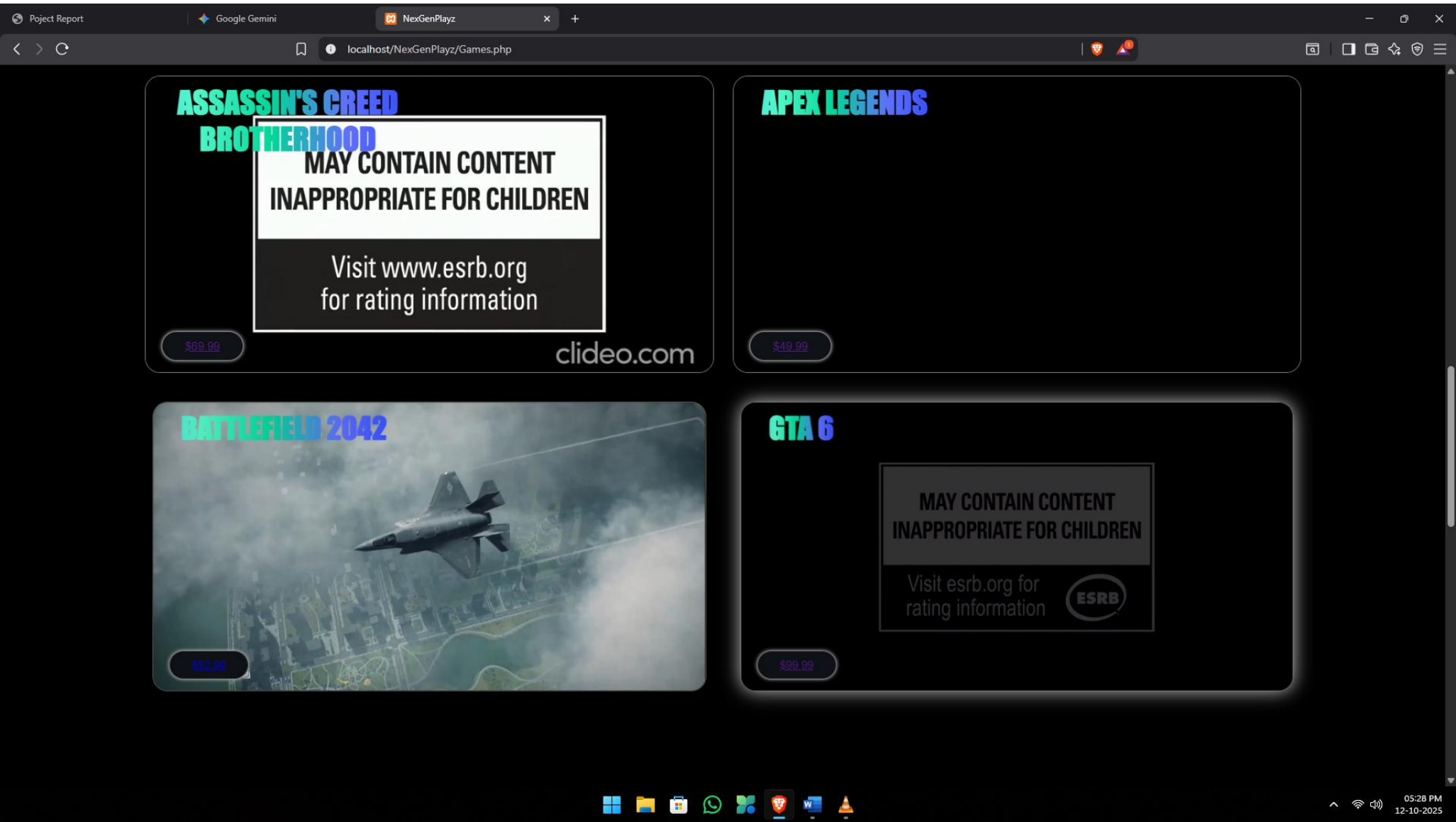
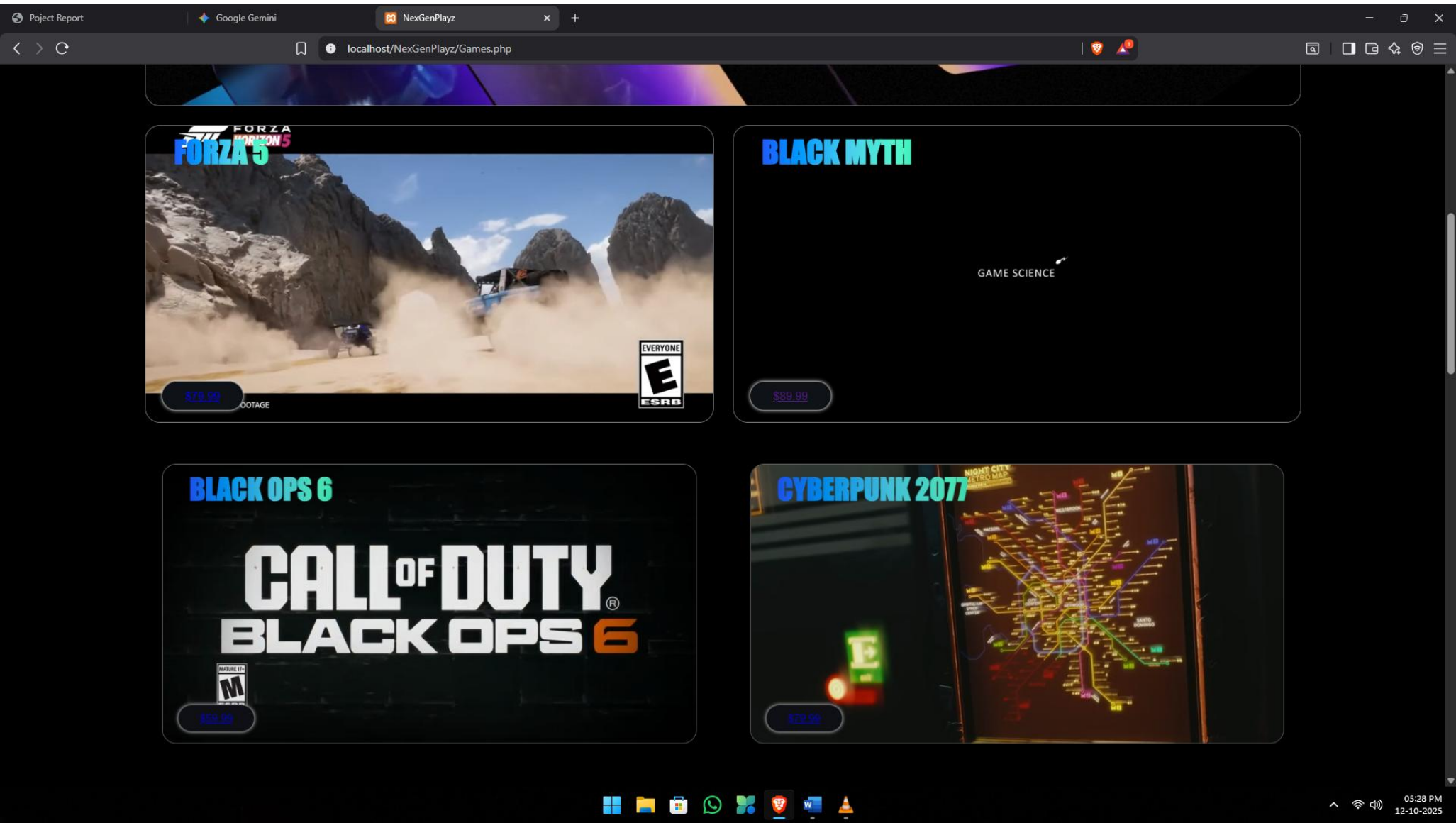
➤ After User Login the User Redirects to Home Page.

4. Game Page:

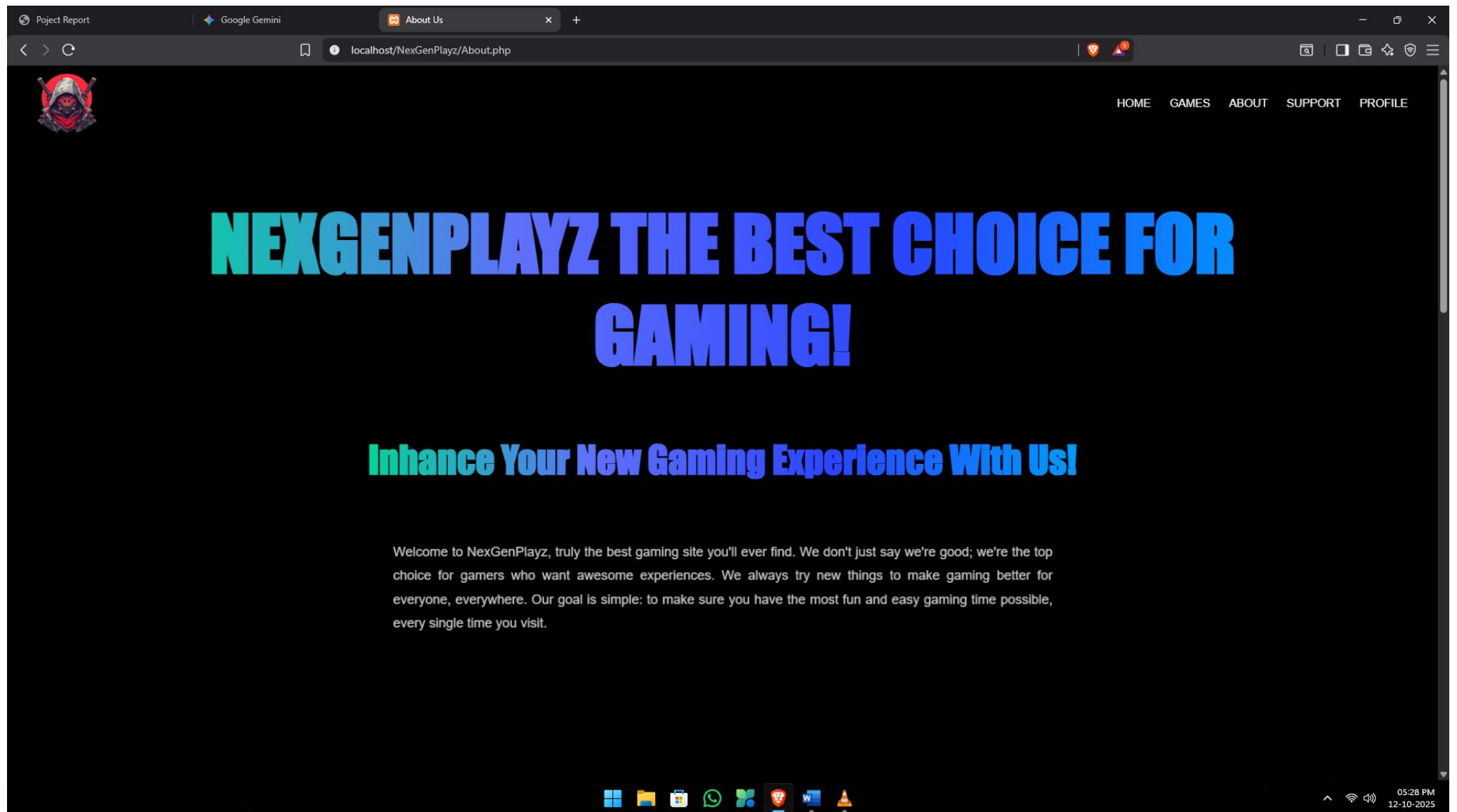


➤ The Game Page Were user can buy games.

5. Game Title and price with intro:

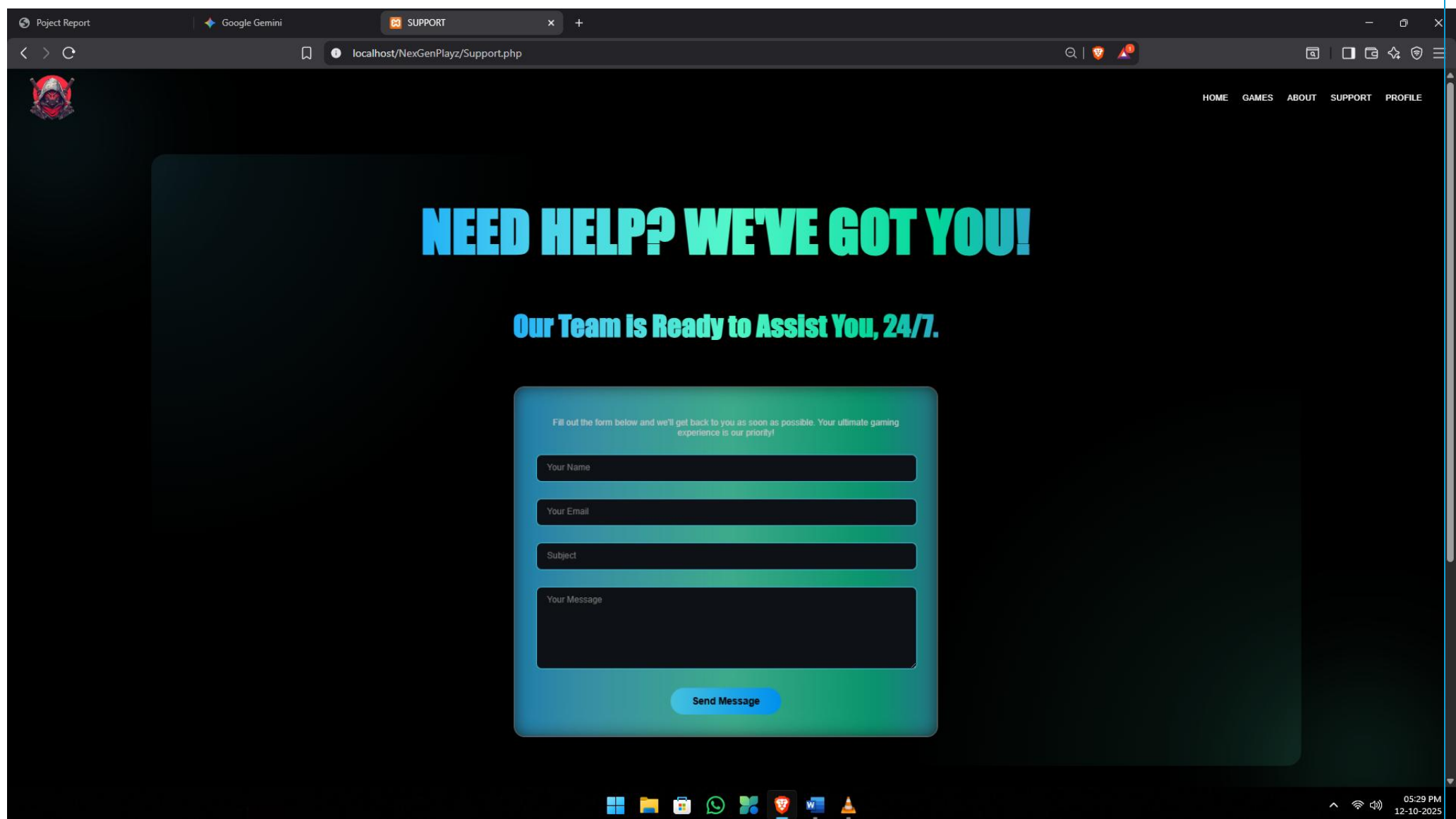


6. About Page:



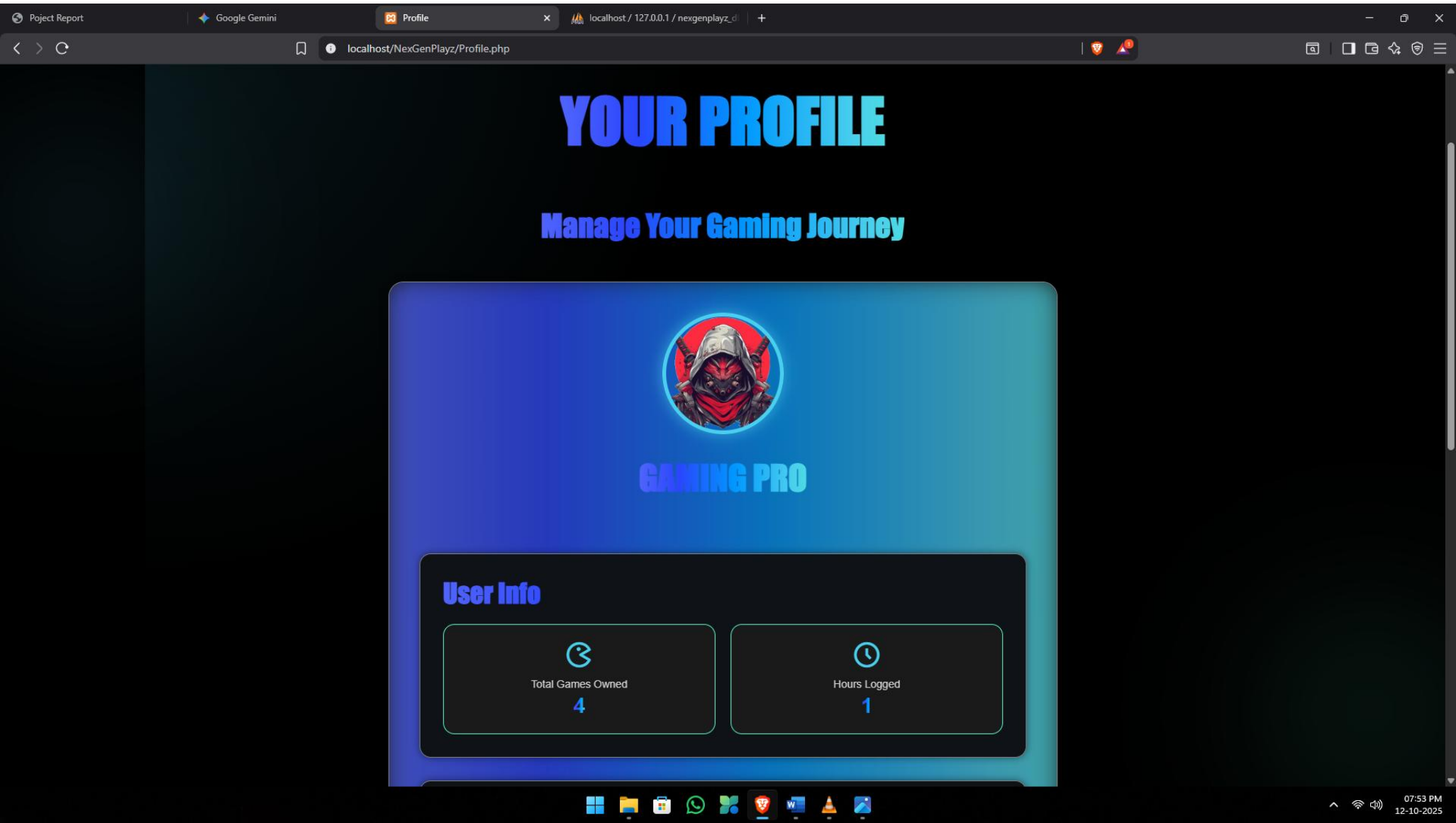
- About Page has our websites history and some best user experiences.

7. Support Page:

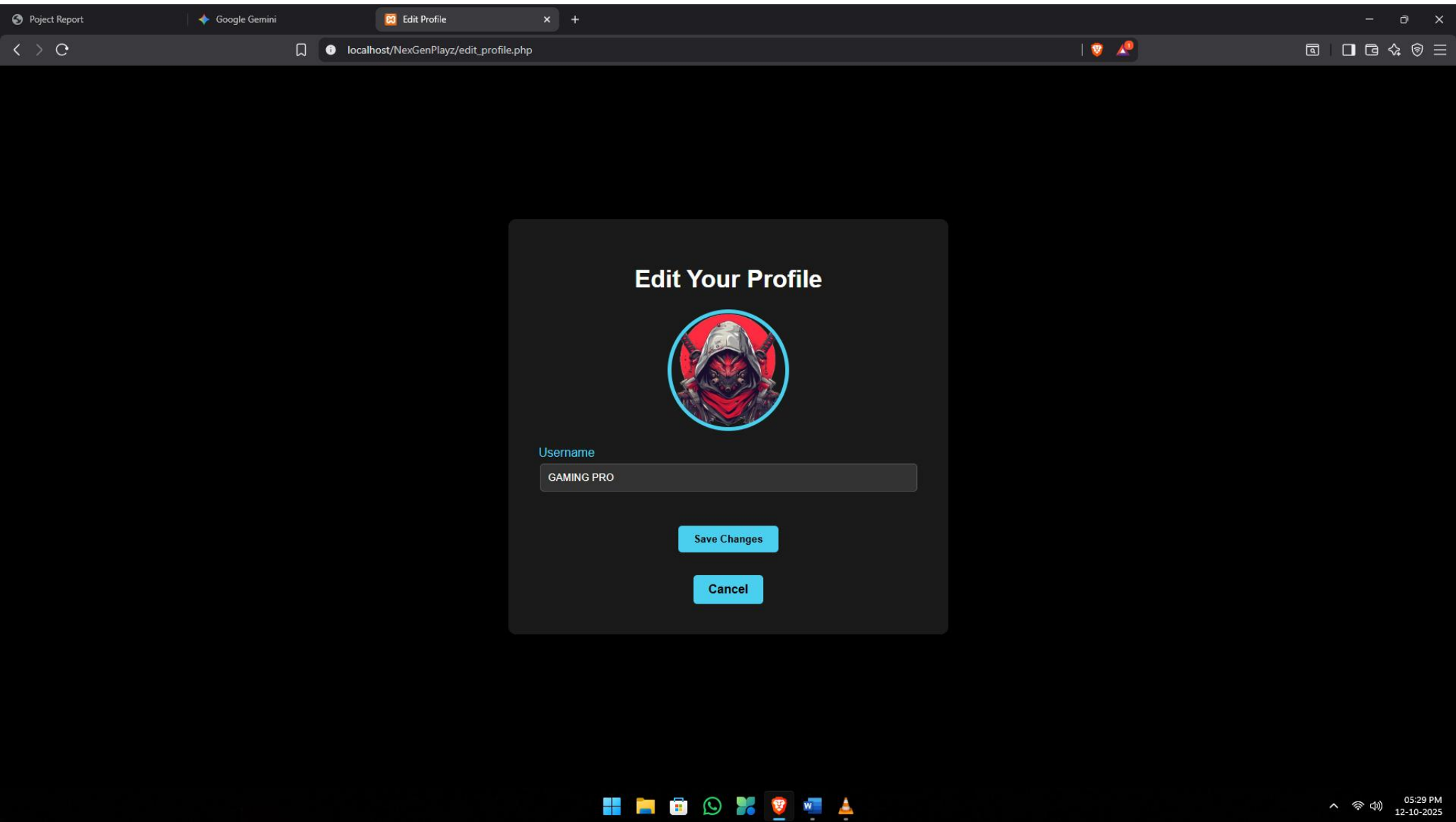


- The support page has form so the user can send any message to us like any issue and feedback.

8. Profile Page:

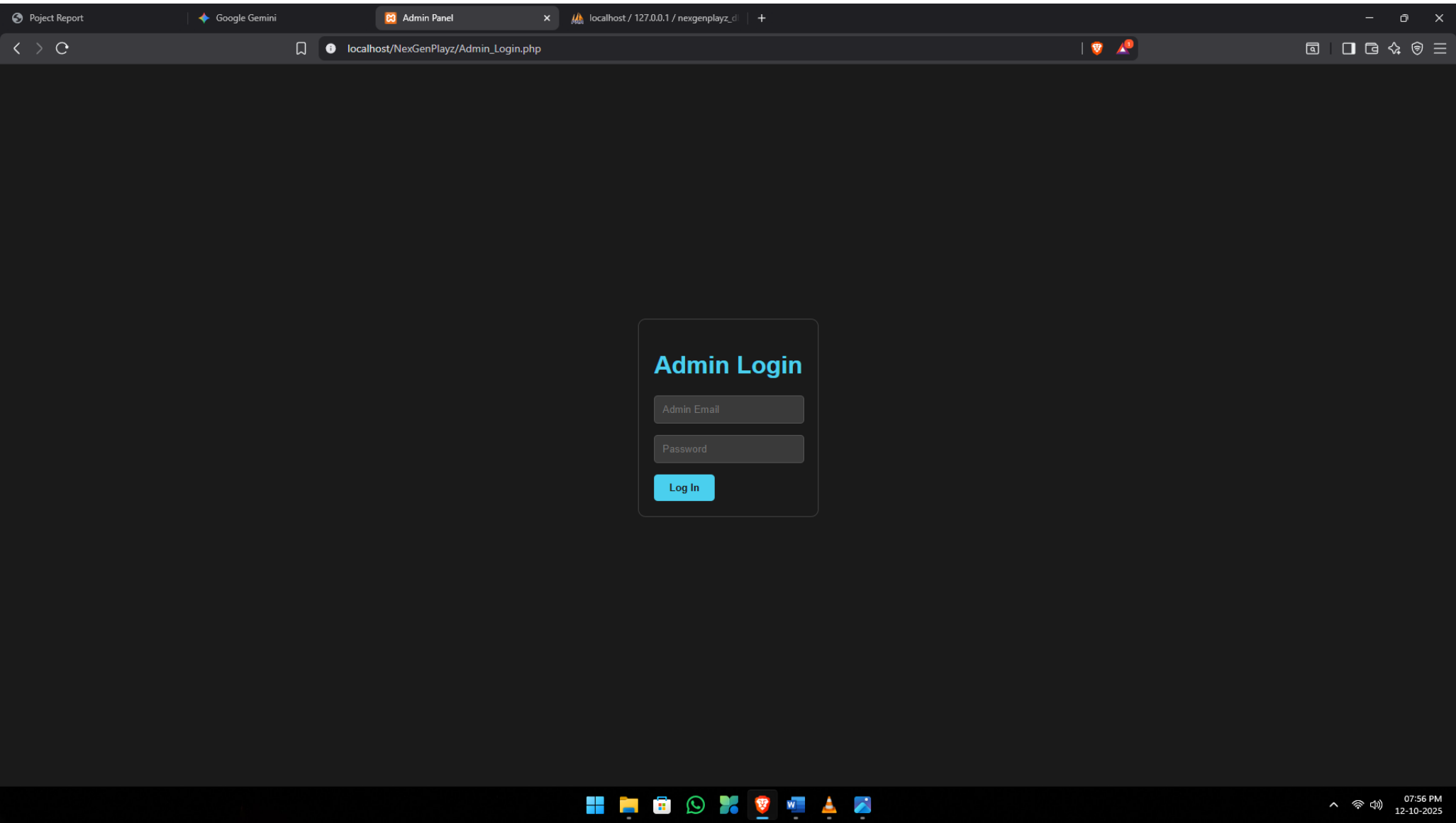


➤ User can see owned games and also edit profile.

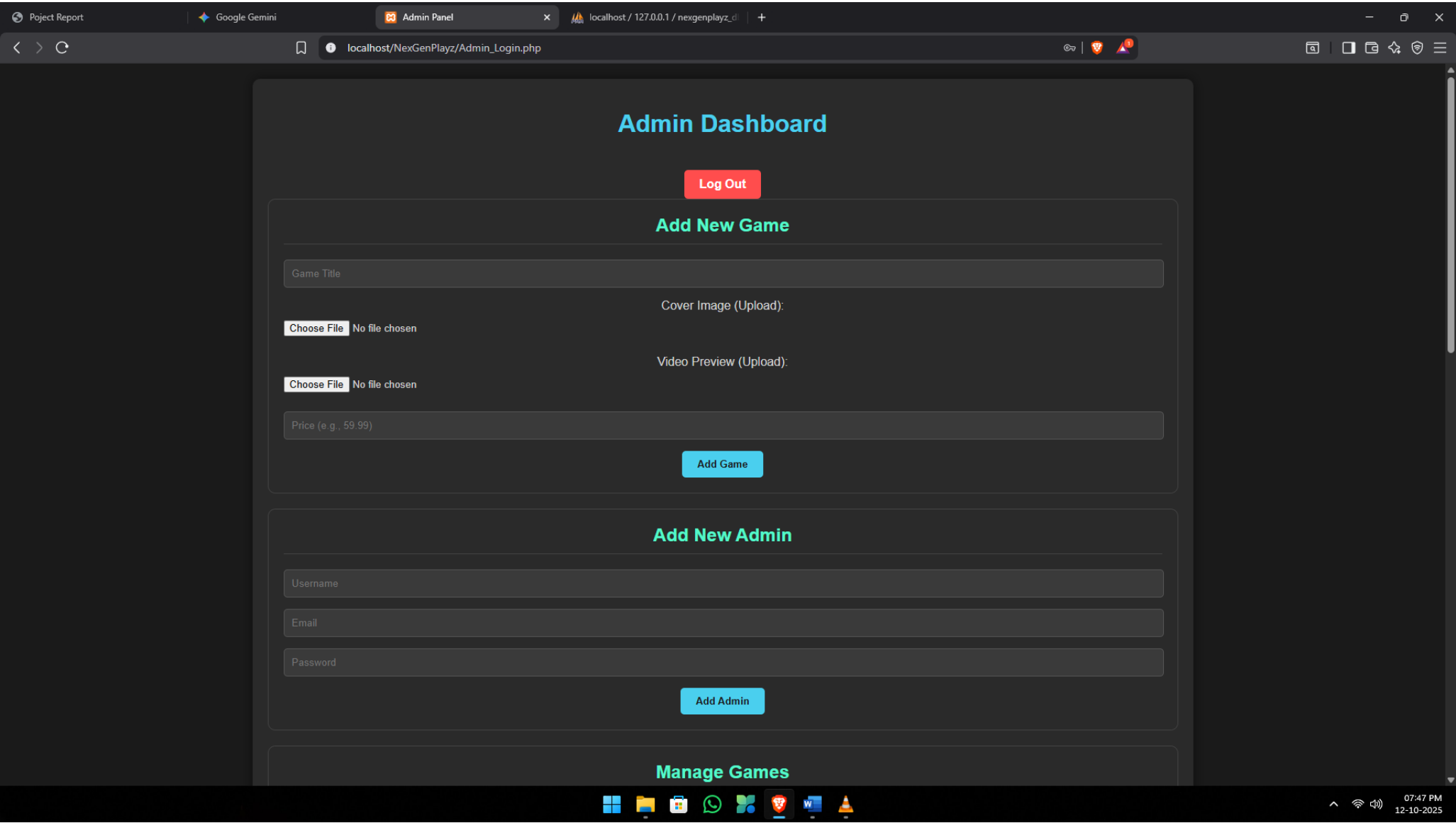


➤ User can change his Profile name and Profile image.

9. Admin Login and Panel:

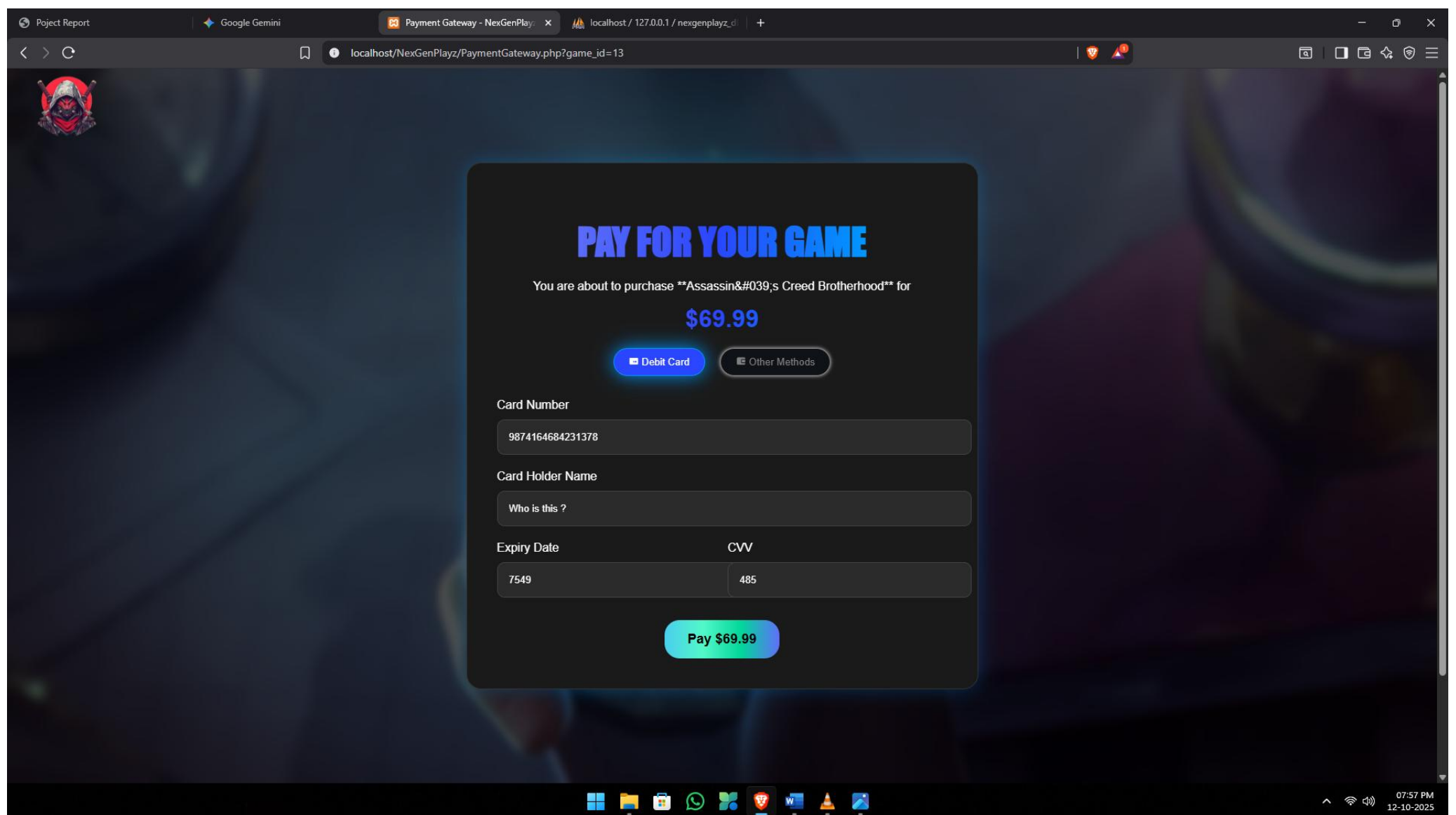


➤ Admin Login page.

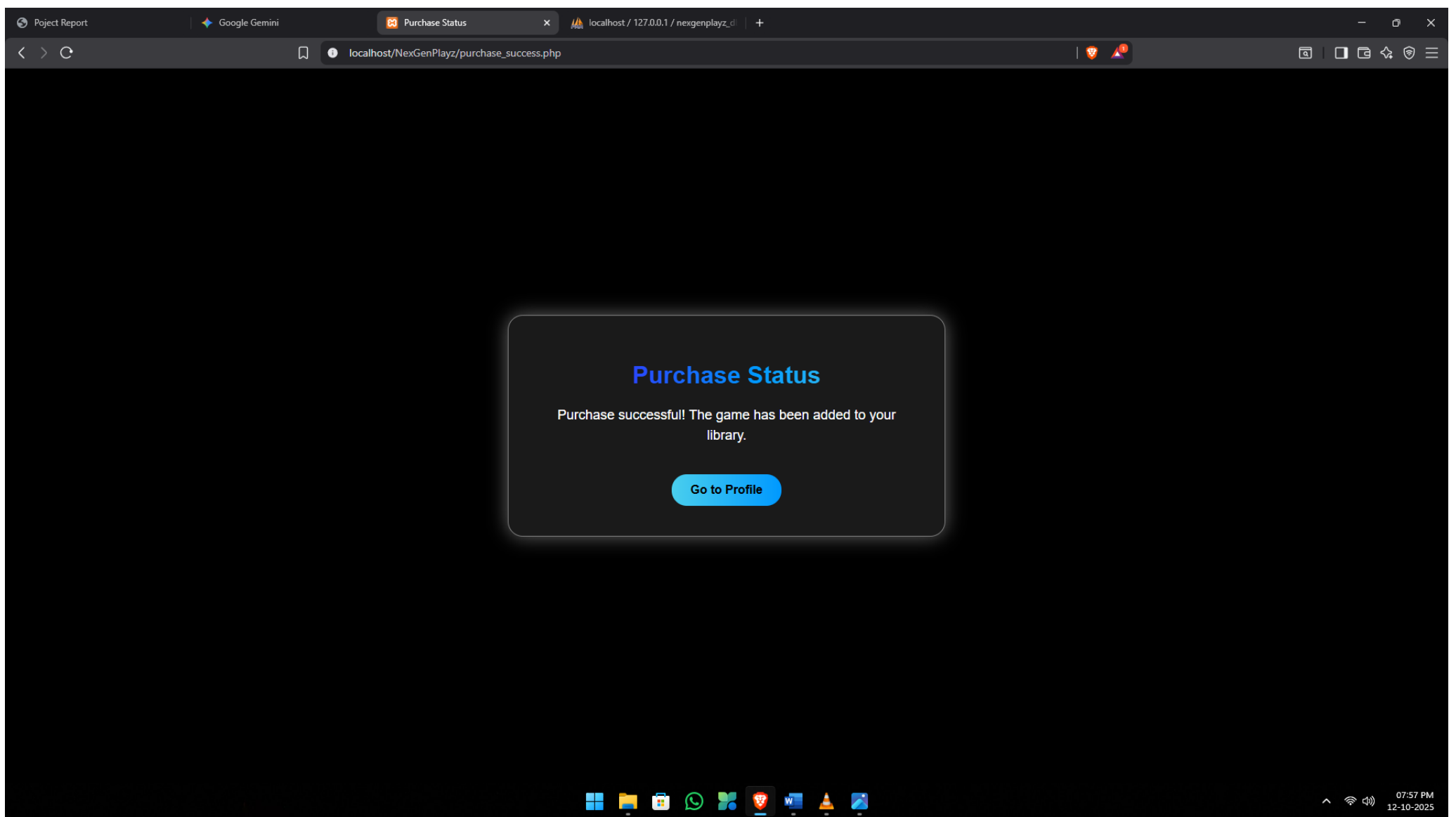


➤ Here Admin can add new games and new admin.

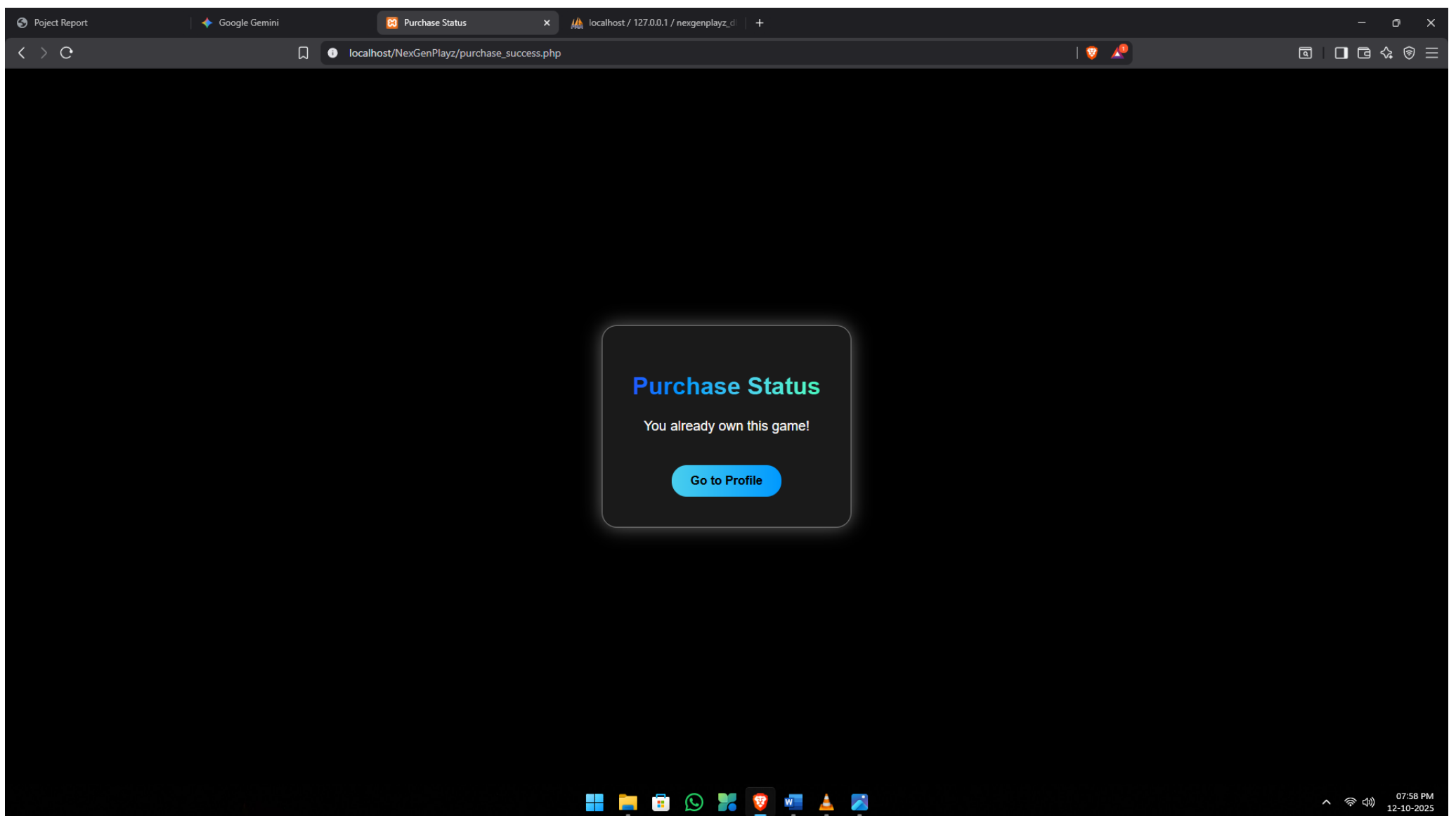
10. Payment and Transaction:



- The payment page that includes game name and price
User need to enter card detail to purchase games.



- If the user purchases the game first time, then this message is displayed.



- If the user has that game, then this message is Displayed.

Limitation

We are trying to make best of the best site but there are some major drawbacks and we need to improve.

- There are limited games.
- Limited payment method.
- If there is so many users than our server can't handle that load.

Future Enhancement

We are known there is so many limitations but we are improving day by day so we can make your experience better and better.

- We are trying to add more game library as many as possible so user have many choices to choose games.
- Improving user experience so the site not have latency and make your experience better.
- We are trying to add more payment options.
- These recognized limitations establish a clear and immediate path for future development. By implementing secure password hashing, integrating a real payment gateway, and introducing features like game catalogue pagination, the NexGenPlayz platform is positioned to evolve into a robust, secure, and user-friendly commercial application.

Conclusion

The NexGenPlayz Web Application successfully meets the core objectives outlined in the project's initial phase, demonstrating a functional, multi-module e-commerce platform for digital game distribution.

Project Successes

1. **Core Functionality Achieved:** The essential modules- **User Authentication**, **Game catalogue** and **User Profile Management** are fully operational. The application correctly manages user sessions, logs user activity time, and restricts unauthorized access.
2. **Database Integration and Data Integrity:** The system utilizes a structured database design and secure prepared statements to handle user records, game inventories, and transactions. The payment simulation successfully handles a mock purchase, updating the user's game ownership record correctly.
3. **Comprehensive SDLC Documentation:** Through the systematic application of the SDLC model, the project has been thoroughly documented, from the initial Data Flow Diagrams (DFD) and Entity-Relationship (E-R) Models to the final **Testing** and **Snapshots** phases, ensuring a verifiable and manageable development process.

Webliography

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- <https://www.w3schools.com/>
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